PERSONALIZED LEARNING PLAN · PREPARED JULY 2026

AI Fluency for Scientific Research

A one-hour-a-day path from AI newcomer to locally-expert practitioner — built for an MD/PhD neuroscientist at the bench.

Foundations → LLMs | Claude & Prompting | Literature Synthesis | No-Code Data Analysis | Scientific Writing | Figures & Slides | Claude Code & Agents

Prepared for	**Ezechukwu “Ez” Nduka — MD/PhD Program, Vanderbilt University**
Stage	**Entering dissertation research · basic-science neuroscience (wet lab)**
Effort budget	**1 focused hour/day · ~6 months**
Spend budget	**≤ $200/month (core path uses a fraction of it)**

Ecosystem-anchored on Anthropic’s Claude, with best-in-class non-Anthropic tools where they win. v1.0

EXECUTIVE SUMMARY

The plan in one page

This plan takes you from “What is AI?” to being the person in your lab others ask for help applying AI to research — in about six months, at one focused hour per day, for well under your $200/month budget.

It is deliberately **hands-on**. Every phase pairs a small amount of high-yield reading, video, or audio with a mandatory exercise done on your **own wet-lab material**: your papers, your protocols, your spreadsheets, your figures, your writing. Fluency here means *doing the task with AI*, not watching someone else do it.

WHAT YOU’LL BE ABLE TO DO

- Explain, in plain language, what an LLM is and how it is trained — and where it fails.
- Run a grounded literature synthesis with **zero fabricated citations**.
- Analyze a real lab dataset conversationally — stats and publication-style plots — without writing code.
- Draft and edit manuscripts and grant text within journal/NIH policy.
- Build figures, schematics, and slide decks fast — ethically.
- Use Claude Code and connectors (PubMed, Drive) to automate research chores.

THE SPEND, DECIDED

Anchor: Claude Pro — $20/mo. Verified to cover nearly everything: Projects, Artifacts, Cowork, Claude in Chrome, and the science connectors (PubMed, Benchling, BioRender, Google Drive).

Add only when a phase needs it: BioRender student plan during figure work; Elicit Plus during a systematic review.

Optional power-up: Claude Max ($100/mo) only if you hit usage limits.

Typical month: **$20–$55**. The $200 cap is comfortable, not a target.

Key findings that shaped this plan

- **The “Anthropic science offering” is real — but the useful part is included in Pro.** Anthropic launched *Claude for Life Sciences* (Oct 2025); its PubMed / Benchling / BioRender connectors work on an ordinary $20 Pro plan. The heavier *Claude Science* workbench skews computational/dry-lab and enterprise, and is **not needed** for you as a non-coding wet-lab scientist. Don’t budget extra for it.
- **Claude is a superb synthesis and writing engine, but it is not a citation database.** The single most important habit in this plan: never let a chatbot invent references — discovery happens in grounded tools (Elicit, Semantic Scholar, Scite, NotebookLM), and every citation is verified in Zotero.
- **Image/video generation is non-Anthropic by necessity.** Claude has none. You’ll use BioRender for real scientific figures and free tools (Google Gemini, Gamma, Canva) for slides — with a hard ethical line: **never generate anything that could pass as experimental data.**
- **No coding required to start.** Because you have no terminal experience yet, Claude Code is deferred to Phase 7 and introduced through a gentle, science-first on-ramp — used to wrangle PDFs and data, not to build software.

HOW “EXPERT” IS MADE CONCRETE

Nine competency domains, each with explicit end-state objectives (Section 2) and a four-level rubric (Novice → Functional → Proficient → Local Expert). Eight milestones (Section 5), one per phase, each a real artifact you produce and self-score — ending in a capstone “research sprint” that chains every skill end to end. Progress is visible, not vibes.

Every resource in this plan was checked against the live web on 8–9 July 2026. Where a specific price or product detail could not be loaded directly (some vendor pages block automated access), it is corroborated from multiple current sources and flagged in Section 10. Consumer-AI prices and model names change monthly — confirm at signup.

CONTENTS

Table of contents

A Appendix: first-week checklist & starter prompts

Copy-paste, wet-lab flavored

HOW TO WORK THROUGH THIS

This document is built to be worked through in order — but it also stands alone as a reference you'll return to. The phases are sequential by design (each assumes the last), while Sections 6–9 are meant to be dipped into anytime. If you ramp faster than expected, the natural place to compress is Phase 1; the high-value core is Phases 3–7. A good rhythm with your mentor: a 15-minute check-in at each milestone (Section 5), using its self-score rubric to keep progress concrete.

SECTION 1

How to use this plan

One focused hour a day is a real constraint, not a soft one. This plan is engineered around it: short inputs, immediate practice, and nothing included that isn't high-yield.

The daily hour

Treat each session as roughly **20 minutes of input** (a video segment, a doc, a short article) and **40 minutes of doing**. The doing is the point. Consuming content about AI produces the *illusion* of fluency; using AI on your own material produces the real thing. Where a resource is long (a 3-hour Karpathy talk), it is explicitly chunked across sessions.

RULES THAT MAKE THE HOUR PAY OFF

- **Always use real lab material.** Your papers, your data, your draft — never a toy example when a real one exists.
- **Keep an AI Lab Notebook.** One running doc: what you tried, the prompt, what worked, what the model got wrong. This becomes your personal playbook.
- **Verify before you trust.** Every fact, number, and citation that leaves a chat and enters your science gets checked against the source.
- **Miss a day, don't miss two.** The schedule has slack built in; consistency beats intensity.

DATA HYGIENE FROM DAY ONE

Your PI confirmed that unpublished basic-science data is generally fine in mainstream cloud AI tools — but build good habits anyway:

- Never paste **human-subjects / identifiable** data into any consumer tool without checking IRB and data-use terms first.
- Strip names, MRNs, and animal-protocol identifiers you don't need.
- Treat anything **pre-publication and competitively sensitive** with care; when unsure, ask your PI.
- Know your setting: consumer Claude does not train on your chats when memory/training controls are off — confirm your account settings.

The shape of the six months

Nine phases (0–8) over ~26 weeks. Each phase has: learning objectives, a curated resource shortlist, a mandatory hands-on exercise tied to your lab, and a milestone you self-score before moving on. The phases build on each other; the rubric in Section 2 lets you (and your mentor) see the climb from novice to local expert in each competency.

PHASE	FOCUS	WEEKS	~HOURS	MILESTONE ARTIFACT
0	Setup & first contact	1 (part)	2–3	Accounts live; first useful chat
1	Foundations: what AI & LLMs are	1–3	12–14	5-minute "explain an LLM" talk
2	Claude fluency & prompting	4–5	9–10	Thesis Project + prompt library

3	Literature review & synthesis	6–9	18–20	Verified mini-synthesis, 0 fake cites
4	Data analysis without coding	10–12	14–15	Real dataset → stats + figure
5	Scientific writing & integrity	13–15	14–15	Edited section + disclosure note
6	Figures, slides & media	16–17	9–10	BioRender figure + AI deck
7	Claude Code, agents & connectors	18–22	22–25	PDF folder → evidence table
8	Integration, capstone & habit	23–26	16–18	End-to-end research sprint

Why six months?

At ~5 sessions/week, six months is roughly **120–130 focused hours** — with real practice on real work. That is the horizon justified by the objectives in Section 2:

- **Shorter (a "30-day crash course") can't reach expertise.** Breadth this wide — foundations, synthesis, analysis, writing, figures, agents — needs spaced practice to stick, and each skill has to be exercised on genuine lab tasks, which take time to arise.
- **Longer than ~8 months works against you.** Tools and model versions drift on a monthly cadence; momentum and retention fade. Six months is long enough for depth and durable habits, short enough that what you learn is still current when you finish.
- **It fits your science calendar.** Phases map onto things a second-to-third-year PhD is actually doing — reading in, generating pilot data, writing a first-year committee document or F30/F31 aims — so the practice is never contrived.

IF LIFE COMPRESSES THE TIMELINE

The plan is modular. A realistic minimum to still call yourself fluent: Phases 0–5 (~10–12 weeks) cover foundations, Claude, literature, data, and writing — the daily-driver skills. Phases 6–8 add figures, agentic automation, and integration, which push you from "fluent" to "the person others ask." Don't skip milestones — they are the proof.

SECTION 2

The destination: what "locally expert" means

"Expert" is easy to say and hard to measure. Here it is defined as nine concrete competency domains, each with end-state objectives you should be able to demonstrate — and a rubric that turns the vague endpoint into a visible climb.

The working definition: **you are locally expert when you can not only do each task below reliably on your own work, but also show a labmate how — and you know the limits and failure modes well enough to keep the lab out of trouble.** Teaching-readiness and risk-awareness are what separate "good user" from "local expert."

The nine competency domains & their objectives

D1 · Foundational literacy. Explain in plain language — to a scientist or a family member — what AI, machine learning, deep learning, and neural networks are; what an LLM is and how it is built (pretraining, fine-tuning, RLHF); and what tokens, context windows, embeddings, parameters, and multimodality mean. Explain *why* LLMs hallucinate and why they're confident when wrong.

D2 · Conversational skill (prompting). Get reliably better output than a novice from the same model: give role/context/constraints, provide examples, ask for structured output, iterate and critique, and know when to switch models or start fresh. Build and reuse a personal prompt library.

D3 · The Claude platform. Use Projects, Artifacts, Cowork, file uploads, Claude in Chrome, and connectors fluently; choose the right surface for a task; set up a persistent Project that carries your thesis context across every chat.

D4 · Literature discovery & synthesis. Run an end-to-end literature workflow — discover with grounded tools, map a subfield, synthesize with Claude/NotebookLM, and manage references in Zotero — producing a synthesis with **no fabricated or misattributed citations**, every claim traceable to a real, read paper.

D5 · Data analysis without coding. Take a real lab dataset (CSV/Excel), analyze it conversationally — cleaning, descriptive stats, appropriate basic tests, and publication-style plots — interpret results critically, catch when the AI does something statistically wrong, and keep the work reproducible.

D6 · Scientific writing & communication. Use AI to outline, draft, tighten, and pressure-test scientific prose (manuscripts, aims, rebuttals, plain-language summaries) while staying squarely within ICMJE and NIH policy — correct disclosure, no undisclosed authorship, no fabrication, no AI in confidential peer review.

D7 · Visual & media generation. Produce clear scientific figures and schematics (BioRender), concept graphics for slides/posters (Gemini, Canva), and AI-generated decks (Gamma) — efficiently and within a firm ethical line that never fabricates data-like imagery.

D8 · Agents & automation (Claude Code + connectors). Comfortably use the terminal at a basic level; run Claude Code to process folders of PDFs/data and automate research chores; understand agents conceptually; connect Claude to PubMed and Google Drive via MCP connectors; know when an agent should *not* be trusted to act unsupervised.

D9 · Responsible use & judgment. Reason clearly about hallucination, bias, privacy, reproducibility, and research integrity; verify AI output as a reflex; explain to others what is and isn't appropriate; make good build-vs-trust decisions. This is the connective tissue across all eight other domains.

The self-assessment rubric

For each domain, you rate yourself against four levels. You start mostly at **Novice**; the plan is designed to carry every domain to **Proficient**, with several reaching **Local Expert** by the capstone. Re-score at each milestone.

LEVEL	WHAT IT LOOKS LIKE (ANY DOMAIN)
Novice	Has heard the terms; uses AI like a search box; accepts output at face value; unaware of failure modes.
Functional	Completes the task with guidance or a template; gets useful results on simple cases; starting to spot obvious errors.
Proficient	Does the task independently on real work; structures prompts and workflows deliberately; verifies output; recovers when the model is wrong.
Local Expert	Fast and reliable on messy real cases; *teaches the workflow to others*; anticipates failure modes and designs around them; makes sound judgment calls about when not to use AI.

TARGET END-STATE PROFILE

By the capstone, a realistic and excellent outcome is **Local Expert** in D1–D4, D6, and D9 (the daily research core), and **Proficient** in D5, D7, and D8 — with clear paths to push those to expert as your own research demands. That profile fully satisfies the goal: the go-to person in your lab for applying AI to science and daily life.

SECTION 3

Tools & budget

The headline: a single **$20/month Claude Pro** subscription is the engine for ~80% of this plan. Everything else is either free or added only for the weeks a phase needs it. The $200 budget is comfortable headroom, not a spending target.

3.1 The Anthropic core

Research confirmed that the surfaces you need are bundled into consumer Claude plans — you do not need the API (pay-per-token, meant for building software) or any enterprise product.

SURFACE	WHAT IT IS	PLAN	WHY IT MATTERS FOR YOU
Claude Pro	The core subscription: full model access, higher limits	**$20/mo** (≈$17/mo annual)	The anchor. Unlocks everything below.
Projects	Persistent workspace with custom instructions + uploaded knowledge every chat inherits	Included	One Project per paper / experiment / grant keeps context loaded.
Artifacts	Live side-panel for documents, tables, small apps	Included	Draft docs, comparison tables, quick calculators.
Cowork	Desktop "AI coworker" that runs multi-step tasks over your files/folders	Included (Pro, since Jan 2026)	"Organize these data files / synthesize these PDFs" over a real folder.
Connectors (MCP)	One-click links to outside tools & data	Included, all plans	**PubMed, Google Drive, BioRender, Benchling** — the science combo.
Claude in Chrome	Browser extension that can act on web pages	Included, paid plans (beta)	Navigate journal portals/databases; use cautiously.
Claude Code	Agentic tool that reads files, runs commands, edits, automates	Included (Pro/Max)	Phase 7: PDF/data wrangling — not software-building.
Claude for Excel	Spreadsheet integration for conversational analysis	Paid plans	Phase 4: analyze CSV/Excel without code.

THE “SCIENCE OFFERING,” ASSESSED HONESTLY

Anthropic really does have a science push — but the practically useful part for you is already inside Pro:

- **Claude for Life Sciences** (launched Oct 2025): connectors into **PubMed, Benchling, BioRender, and 10x Genomics**, plus science “Agent Skills.” These work on a normal Pro plan — *this is the real, budget-friendly win*, and it maps directly onto your wet-lab neuro workflow (literature, protocols, figures).
- **Claude Science** (the newer “AI workbench”): a desktop app oriented toward computational pipelines, packages, and compute, in macOS/Linux beta and leaning enterprise/dry-lab. **Verdict: overkill and not needed** for you as a non-coding wet-lab scientist. Don’t budget for it; revisit only if your research turns computational.

Some product/pricing specifics here were corroborated from official-page search snippets rather than loaded directly (vendor bot-blocking) — see Section 10. None affect the recommendation: start on Pro.

3.2 When (and whether) to go beyond Pro

Claude Max ($100/mo for 5× usage; $200/mo for 20×) buys more usage and priority, not different core features. Start on **Pro** and only step up to Max if you repeatedly hit limits during heavy synthesis or data-analysis days — likely around Phases 3–4 or 7, if at all. It is a usage upgrade, not a prerequisite.

Check first, might be free: Vanderbilt may participate in **Claude for Education** or provide institutional access to paid tools (e.g., Scite via the library, Microsoft 365 Copilot). A five-minute check with your library / IT could make some of this free. (No public self-serve .edu discount for consumer Claude existed as of mid-2026 — the education path is institution-licensed.)

3.3 The full stack & a realistic monthly budget

LAYER	TOOL	COST	WHEN
AI assistant, synthesis, writing, analysis, agents	**Claude Pro**	$20/mo	Always — month 1 on
Reference manager & citation gatekeeper	**Zotero**	Free	From Phase 3
Grounded discovery & subfield mapping	**Semantic Scholar**, **ResearchRabbit**	Free	From Phase 3
Grounded close-reading + audio overviews	**NotebookLM** (free tier)	Free	From Phase 3
Evidence tables from real papers	**Elicit** (free tier, or Plus)	$0–12/mo	Phase 3, systematic-review weeks
“Did this finding hold up?”	**Scite** (check library first)	$0–12/mo	Optional, Phase 3
Scientific figures & schematics	**BioRender** (academic/student)	~$35/mo	Phase 6 & when making figures
Free diagramming	**draw.io**, **Excalidraw**	Free	Anytime
AI images for slides	**Google Gemini** (free tier)	Free	Phase 6

AI slide decks	**Gamma** (free, or Plus $8–10)	$0–10/mo	Phase 6
Design/posters + Claude connector	**Canva** (free / Education free)	Free*	Phase 6
Optional usage upgrade	Claude Max	+$100/mo	Only if hitting limits

*Canva for Education is free if eligible; otherwise a free personal tier covers basic needs. Prices are academic/student rates where available and were current as of July 2026 — confirm at signup with a .edu email.

TYPICAL MONTH

$20 (Pro alone) for most months. A figure-heavy or systematic-review month might reach **$35–$55** with BioRender + Elicit added. Even stacking Max on top stays under **$160** — below the cap with room to spare.

THE ONE-LINE SPENDING RULE

Pay for Claude Pro always. Add a second paid tool *only* in the phase that uses it, and cancel it when that phase ends. Everything else in this plan is genuinely free.

SECTION 4

The phased timeline

Nine phases over ~26 weeks. Each lists its objectives, a short verified resource set, a mandatory hands-on exercise on your own lab material, and the milestone that closes it. Every resource name is a live link; costs are consolidated in Section 7.

HOW TO READ EACH PHASE

Learn = the ~20-minute inputs. **Do** = the ~40-minute practice, always on real work. **Milestone** = the artifact you produce and self-score (Section 5) before advancing. Resource names in **bold** are clickable and also detailed in Section 7.

PHASE 0 WEEK 1 (PART) · 2–3 HRS · DOMAINS D3, D9

Setup & first contact

The goal of this short phase is zero friction: accounts live, tools installed, and one genuinely useful interaction — so momentum starts on day one. No terminal, no coding.

Do

- Create/upgrade to **Claude Pro** ($20/mo). Install the **Claude desktop app** (Mac or Windows) and try **claude.ai** in the browser — no command line involved.
- In account settings, review privacy/memory/training controls and set them to your comfort (see data-hygiene box, Section 1).
- Make sure you have a **Google account** (for the free NotebookLM and Gemini used later).
- Start your **AI Lab Notebook** (a single Google Doc or Claude Project) — every session's notes go here.
- **First useful chat:** paste the abstract of a paper you're currently reading and ask Claude to explain the key finding, define three unfamiliar terms, and list two questions a skeptical reviewer would ask.

Learn (light)

- Skim the **Anthropic Academy** catalog and bookmark it — free official courses you'll use in Phases 2 and 7.

✦ Checkpoint 0

Accounts are live, the desktop app opens, and you've had one chat that actually helped you understand a paper. Your AI Lab Notebook exists with its first entry.

PHASE 1 WEEKS 1–3 · 12–14 HRS · DOMAINS D1, D9

Foundations: what AI & LLMs really are

Before using the tools deeply, you'll build an honest mental model of what's under the hood. As a neuroscientist you'll find the neural-network framing intuitive — and understanding *why* these systems hallucinate is what makes you a safe, critical user forever after.

Objectives

Explain AI vs ML vs deep learning; what a neural network and an LLM are; pretraining, fine-tuning, and RLHF; tokens, context windows, embeddings, parameters; and the mechanistic reason hallucinations happen.

Learn (chunk the long videos across sessions)

- **3Blue1Brown** — "But what is a neural network?" (Ch.1, ~19 min) and the short "Large language models explained briefly" (~8 min). The best visual intuition anywhere; watch the transformer & attention chapters (Ch.5–6) later in the phase.
- **Andrej Karpathy — "[1hr Talk] Intro to Large Language Models"** (~1 hr, split over 2–3 sessions). The single best conceptual map of training and the "LLM as OS/agent" idea.
- **Tim Lee & Sean Trott — "Large language models, explained with a minimum of math and jargon"** (long read). A cognitive scientist co-authored it, so the framing lands well for a neuroscientist — embeddings, transformers, training.
- **Structured anchor (pick one, run through the phase): Elements of AI** (Helsinki, free) for a no-math "what is AI" spine, *or* **Generative AI for Everyone** (DeepLearning.AI, free audit) for the modern LLM-era version.

Do

- After each video, write a 3-sentence explainer *in your own words* in your AI Lab Notebook. Teaching-back is the retention trick.
- Ask Claude to quiz you on the day's concept, then to grade your answers and correct misconceptions.
- Deliberately **catch a hallucination**: ask Claude for five references on a niche neuro topic, then check whether they exist (they may not). This lesson underwrites all of Phase 3.

✦ Milestone 1 — "Explain an LLM"

Record a **5-minute talk** (phone camera or slides) explaining to a labmate what an LLM is, how it's trained, and why it hallucinates — no notes. Self-score D1 on the rubric. *Local-expert tell:* you can field "so is it just autocomplete?" without hand-waving.

PHASE 2 WEEKS 4–5 · 9–10 HRS · DOMAINS D2, D3

Claude fluency & prompting

Now you turn the mental model into skill: getting consistently strong output, and setting up Claude to carry your research context. This is where the daily productivity gains begin.

Objectives

Prompt deliberately (role, context, constraints, examples, structured output, iteration); use Projects, Artifacts, and Cowork; choose the right surface for a task; build a reusable prompt library.

Learn

- **Anthropic Prompt Engineering Interactive Tutorial** (official, free) — work the concepts; as a non-coder you can skip the code cells and keep the prompting lessons.
- **Anthropic Academy** prompting course (free, official) for a structured pass with a completion certificate.
- **Andrej Karpathy — "How I Use LLMs"** (practical companion; tool use, web search, thinking models) — chunk across sessions.

Do (all on real lab work)

- Create a **Claude Project** for your thesis area: add custom instructions (your subfield, techniques, writing style) and upload 5–10 core papers and your lab's key protocols as project knowledge.
- Start a **prompt library** in your AI Lab Notebook: reusable prompts for "summarize this paper for my committee," "critique this paragraph," "turn these bullet points into methods prose," "explain this technique to a rotation student."
- Use an **Artifact** to build a comparison table of three methods you're choosing between.
- Try **Cowork** on a folder: point it at a set of PDFs and ask for a one-paragraph summary of each.

✦ Milestone 2 — Thesis Project + prompt library

A working Claude Project loaded with your research context, plus **10+ reusable prompts** you actually use. Demonstrate by getting a genuinely useful, well-structured answer to a real research question in a single well-formed prompt. Self-score D2, D3.

PHASE 3 WEEKS 6–9 · 18–20 HRS · DOMAINS D4, D9

Literature review & synthesis

The highest-leverage research skill in the plan — and the one with the sharpest integrity risk. The core discipline: **discovery happens in grounded tools that search real papers; Claude and NotebookLM synthesize only what you have actually retrieved; Zotero is the gatekeeper for every citation.**

Objectives

Run an end-to-end literature workflow — discover, map, synthesize, and manage references — producing a synthesis with no fabricated or misattributed citations, every claim traceable to a real paper you have read.

Learn the tools (each <1 hr; use verified tutorials in Section 7)

TOOL	ROLE IN THE WORKFLOW	COST
Zotero	Reference manager & citation gatekeeper — only real, retrieved papers live here	Free
Semantic Scholar	Free search backbone + TLDRs + alerts	Free
ResearchRabbit	Visual citation-network mapping of a subfield; syncs to Zotero	Free
Elicit	Structured evidence tables from real papers (methods, n, outcomes)	Free / $12
Consensus	Fast citation-backed check on yes/no empirical claims	Free / ~$10
Scite	Did a finding hold up? Supporting vs contrasting citations	Library / $12
NotebookLM	Grounded close-reading of papers you upload; cites every claim; audio overviews	Free
Claude (PubMed connector)	Synthesis, drafting, and reasoning across retrieved papers	In Pro

Do

- Pick a **real question** from your project (e.g., “what’s known about [receptor/pathway] in [cell type] during [process]”).
- **Discover:** map the subfield in ResearchRabbit, pull an evidence table in Elicit, sanity-check claims in Consensus, and check whether key findings replicated in Scite. Import everything real into **Zotero**.
- **Synthesize:** load the retrieved PDFs into **NotebookLM** and a **Claude Project**; ask each to summarize themes, agreements, and open questions — every claim cited to a source you have.
- **Verify:** for every citation, confirm it exists and says what’s claimed (quote-search in the PDF / Google Scholar; check DOIs via Crossref; screen older references against **Retraction Watch**).

THE NON-NEGOTIABLE RULE

Never ask Claude (or any chatbot) to “give me references for X” from memory — it will produce real-looking citations that don’t exist. Fabricated-reference rates in the literature have risen sharply. Ask it to reason only over papers you supply. Zotero is the single source of truth for your bibliography.

✦ Milestone 3 — Verified mini-synthesis

A **2–3 page synthesis** on a real question, ~10–15 papers, with a Zotero library and a short **verification log** proving every citation was checked — **zero fabricated or misattributed references**. This is the plan’s integrity keystone. Self-score D4, D9. *Local-expert tell:* you can teach the “discover-then-synthesize, never-invent” workflow to a labmate.

PHASE 4 WEEKS 10–12 · 14–15 HRS · DOMAINS D5, D9

Data analysis without coding

You have no coding background — and won't need one. Modern AI tools write and run the analysis code invisibly; you work entirely in plain English. The skill to build is **directing and checking** the analysis, not writing it.

Objectives

Take a real CSV/Excel dataset, analyze it conversationally — clean, describe, run appropriate basic tests, and make publication-style plots — interpret critically, catch statistical mistakes, and keep it reproducible.

Learn

- **Claude for Excel** (official support guide) — your primary path, since Claude is your main tool.
- **OpenAI — "Data analysis with ChatGPT"** (official help doc) — a clear reference for the upload-and-ask pattern (optional; only if you also keep a ChatGPT Plus seat).
- **Coupler.io — "Use Claude.ai for Data Analytics (Safely)"** — especially the privacy section: what not to paste.
- *Preview of Phase 7:* **VelvetShark — "Data analysis with Claude Code"** for datasets too big for chat upload.

Do (on a real dataset from your rotation/lab)

- Upload a real spreadsheet; ask Claude to describe its structure, flag data-quality issues, and propose a cleaning plan — then approve each step.
- Run **descriptive stats** and generate a clear plot (bar with error bars, scatter, or time-course as fits your data).
- Ask which statistical test is appropriate and **why** — then have it check assumptions (normality, sample size, independence). *Deliberately probe:* ask a leading question and confirm it pushes back rather than agreeing.
- Have it produce a short, plain-language interpretation you could show your PI — and independently sanity-check one number by hand.

STATISTICS GUARDRAIL

AI will confidently run the *wrong* test if asked leading questions. Treat it as a fast analyst who needs supervision: make it justify each test, state assumptions, and show its work. For anything headed toward publication, a real statistician or your PI still signs off. Never upload identifiable human data without IRB/data-use clearance.

✦ Milestone 4 — Real dataset → stats + figure

Start from a raw lab spreadsheet and produce a **cleaned dataset, an appropriate statistical result you can defend, and a publication-style figure** — documenting the steps so it's reproducible. Self-score D5. *Tell:* you caught at least one thing the AI got wrong or oversimplified.

PHASE 5 WEEKS 13–15 · 14–15 HRS · DOMAINS D6, D9

Scientific writing & research integrity

Writing is where AI saves the most time and carries the most policy risk. You'll learn to use it as a tireless editor and sparring partner — firmly inside the rules that govern manuscripts and grants.

Objectives

Use AI to outline, draft from your own notes, tighten, and pressure-test scientific prose (manuscript sections, aims, rebuttals, plain-language summaries) — with correct disclosure and no policy violations.

Learn (read the policy first — it's load-bearing)

- **ICMJE Recommendations** — AI cannot be an author; all AI use in writing/editing must be disclosed; authors are fully responsible for every fact and citation; reviewers must not upload manuscripts to AI tools.
- **NIH NOT-OD-23-149** — generative AI is prohibited in peer review (uploading applications breaches confidentiality).
- **NIH "Apply Responsibly" (2025)** — applications substantially developed by AI aren't considered original; directly relevant to your future F30/F31 and grant writing.
- **ACS Nano best-practices** and the **MDPI good-practices** checklist — concrete OK-vs-not-OK examples.

APPROPRIATE

- Grammar, clarity, flow, and structure edits on *your own* draft
- Reordering and tightening a rebuttal letter you wrote
- Plain-language summaries and outlines
- "What would a reviewer attack here?" critique
- Disclosed use, per journal policy

NOT APPROPRIATE

- Generating substantive content and passing it off as yours
- Any fabricated data, figures, or citations
- Undisclosed use where disclosure is required
- Uploading confidential manuscripts under review
- Any AI use in NIH peer review

Do

- Take a **real paragraph you wrote** and iterate it with Claude for clarity — keeping your voice and every claim your own.
- Draft a **Specific Aims** outline or a **methods section** from your own bullet points; have Claude critique it as a study-section reviewer.
- Draft a **response-to-reviewers** letter for a paper (yours or a mentor's, with permission).
- Write a proper **AI-use disclosure statement** for a manuscript.

✦ Milestone 5 — Edited section + disclosure

A piece of real scientific writing you **materially improved with AI while staying within policy**, plus a correct disclosure statement and a one-paragraph note on where you drew the line. Self-score D6, D9.

PHASE 6 WEEKS 16–17 · 9–10 HRS · DOMAINS D7, D9

Figures, slides & media

Claude has no image or video generation, so this phase is deliberately non-Anthropic — used directly or through the Canva–Claude connector. The skill is producing clear scientific visuals fast, inside a hard ethical line.

Objectives

Make publication-grade schematics (BioRender), concept graphics for slides/posters (Gemini, Canva), and AI-generated decks (Gamma) — efficiently and without ever fabricating data-like imagery.

THE BRIGHT LINE (APPLIES TO EVERYTHING HERE)

Never use any image or video tool to generate or "clean up" anything that could pass as experimental data — blots, micrographs, histology, flow plots, IHC. That is fabrication, journals screen for it, and it ends careers. AI imagery is for **conceptual illustration and slide graphics only**, disclosed per journal policy.

Learn & Do

- **BioRender** (student/academic plan) with its official Learning Hub — build a **real schematic** of your experimental design or a signaling pathway in your system. Its icons are vetted for scientific accuracy; general image AI can't match that. (For non-biological flowcharts, free **draw.io**/**Excalidraw** work fine.)
- **Google Gemini** (free) — generate a concept/background graphic for a title slide. Strong photorealism, best free access.
- **Gamma** (free, or Plus ~$8–10) — turn your Milestone-3 synthesis or an outline into a **lab-meeting deck** in minutes, then refine.
- **Canva** (free / Education) — optionally connect the verified **Canva MCP connector** to Claude and drive a poster/slide design from a chat.

✦ Milestone 6 — Figure + deck

One **BioRender schematic** good enough for a real talk or paper, and one **AI-generated slide deck** for an actual lab meeting — produced in a fraction of your usual time. Self-score D7. *Tell:* you can state exactly why none of it crosses the data-fabrication line.

PHASE 7 WEEKS 18–22 · 22–25 HRS · DOMAINS D8, D9

Claude Code, agents & connectors

The most ambitious phase, and the reason it's five weeks: you've never used a terminal. The on-ramp is gentle and science-first — Claude Code as a tool that reads a folder of files and does tedious work, *not* as a way to build software.

Objectives

Use the terminal at a basic level; run Claude Code to process folders of PDFs/data and automate chores; understand agents conceptually; connect Claude to PubMed and Google Drive via MCP; and know when an agent should not act unsupervised.

Week-by-week (gentlest → most capable)

WK	FOCUS	KEY RESOURCES
18	Terminal primer — open it, navigate folders, run a command	**Anthropic Terminal guide**; **MDN command-line** (§1–3)
19	What an agent *is* (chatbot answers; agent acts, in a loop)	**IBM "What are AI Agents?"**; **explainX** beginner guide
20	Install & first Claude Code session (substitute science tasks for the code examples)	**Claude Code Quickstart**; **CC for Everyone — Non-Developers**
21	Claude Code for research: a folder of PDFs → themes & evidence tables	**CC for Everyone — Researchers**; **Academy: Claude Code in Action**
22	MCP & connectors: plug Claude into PubMed & Google Drive	**MCP intro** + **modelcontextprotocol.io**

Do

- Complete the terminal primer until "open a terminal, go to a folder, run one command" feels routine. (On Windows, Anthropic's native installer works — WSL is not required.)
- Install Claude Code; in your first session, ask it to summarize a single PDF in a folder.
- **The signature exercise:** point Claude Code at a folder of 10–20 papers/notes and have it produce a cross-file **evidence table or themed summary** with source filenames and verbatim quotes. (Note practical limits: scan-only PDFs need OCR first; long PDFs are read in chunks.)
- Enable the **PubMed** and **Google Drive** connectors in Claude and run one real task through each.

REFERENCE, NOT HOMEWORK

Anthropic's **"Building effective agents"** is the canonical piece, but it's written for engineers. Skim only the opening definitions and the "workflows vs agents" distinction — and save the rest for if you ever go deeper.

✦ Milestone 7 — PDF folder → evidence table

Using Claude Code, turn a folder of real papers into a **useful cross-document artifact** (themed synthesis or evidence table with sources), and connect at least **one MCP connector** (PubMed or Drive) to a real task. Self-score D8. *Tell:* you can explain when you'd *not* let the agent act without review.

PHASE 8 WEEKS 23–26 · 16–18 HRS · ALL DOMAINS

Integration, capstone & the keep-current habit

The final phase chains every skill into one real research workflow, consolidates judgment, and sets up the lightweight habit that keeps you current after the plan ends.

Learn (consolidate judgment)

- **Ethan Mollick — *Co-Intelligence*** (read across the phase): the best narrative frame for working *with* AI as coworker, tutor, and coach — and for everyday/personal use (email, planning, learning).
- **MIT Sloan — "When AI Gets It Wrong"** and **AI Snake Oil / normaltech.ai**: the skeptical, integrity-minded counterweight to hype every scientist should carry.
- Set up your **continuing-education habit** from Section 9 — two podcasts, two newsletters, no more.

Do — the capstone

Choose a real, small research question and run it **end to end, AI-accelerated**, documenting each step and every verification:

1. **Discover & synthesize** the relevant literature (Phase 3 workflow), citations verified.
2. **Analyze** a relevant dataset conversationally and make a figure (Phase 4).
3. **Build** a schematic and a short deck (Phase 6).
4. **Write** a 1–2 page brief with AI editing, within policy, with a disclosure note (Phase 5).
5. **Automate** one chore in the process with Claude Code or a connector (Phase 7).
6. Keep an **integrity/verification log** throughout (Phase 9 judgment).

✦ Milestone 8 — Capstone research sprint

A single documented workflow that takes a real question from literature to a written, figure-supported brief — every AI use appropriate, every citation and number verified. Re-score **all nine domains**. Meeting the target profile (Section 2) = locally expert. *Final tell:* a labmate could follow your write-up and reproduce the workflow.

AFTER THE PLAN

Fluency decays without use and tools drift fast. Keep sharp by (1) using these workflows in your actual research daily, (2) the Section 9 continuing-ed habit, and (3) revisiting your rubric each semester to push D5/D7/D8 toward expert as your project demands.

SECTION 5

Milestones & checkpoints

Eight milestones, one per phase, each a real artifact you produce and self-score — so "progress toward expert" is a tracker you fill in, not a feeling. Print this page; use it at each mentor check-in.

#	MILESTONE ARTIFACT	DOMAINS	"DONE" LOOKS LIKE	SELF-SCORE (N / F / P / LE)
0	Setup & first useful chat	D3	Accounts live; one chat that helped	—
1	5-min "explain an LLM" talk	D1, D9	Explains training + hallucination, no notes	D1 ___ D9 ___
2	Thesis Project + 10-prompt library	D2, D3	One prompt → genuinely useful answer	D2 ___ D3 ___
3	Verified mini-synthesis (10–15 papers)	D4, D9	**Zero fabricated citations**; verification log	D4 ___ D9 ___
4	Real dataset → stats + figure	D5, D9	Defensible test; caught an AI error	D5 ___ D9 ___
5	AI-edited section + disclosure	D6, D9	Improved & within ICMJE/NIH policy	D6 ___ D9 ___
6	BioRender figure + AI deck	D7	Talk-ready; no data-like fabrication	D7 ___
7	PDF folder → evidence table + connector	D8, D9	Cross-doc artifact; 1 MCP connector live	D8 ___ D9 ___
8	Capstone research sprint (end-to-end)	All	Reproducible; every use appropriate & verified	All ___

N = Novice, F = Functional, P = Proficient, LE = Local Expert (rubric, Section 2). Target end-state: LE in D1–D4, D6, D9; P in D5, D7, D8.

FAST CHECKPOINTS BETWEEN MILESTONES

- **Weekly (2 min):** did I do all my sessions on *real* lab material? Is my AI Lab Notebook current?
- **Per phase:** can I explain this phase's skill to a labmate in 3 minutes?
- **The verification reflex:** did anything leave a chat and enter my science unchecked this week? (Target: never.)

WHAT TIPS A DOMAIN INTO "LOCAL EXPERT"

Two signals, both observable: (1) you **teach the workflow** to someone else and they succeed with it; and (2) you **predict the failure mode** before it happens and design around it. When both are true for a domain, mark it LE.

SECTION 6 · DELIVERABLE 2

Glossary of AI terms

Plain-English definitions, grouped so you can learn them in the order the plan introduces them. Skim it in Phase 1; return to it whenever a term appears.

Foundations

Artificial intelligence (AI)

The broad field of building software that performs tasks we associate with human intelligence — recognizing patterns, using language, making decisions. An umbrella term, not one technology.

Machine learning (ML)

A branch of AI where systems *learn patterns from data* instead of following hand-written rules. Show it thousands of examples and it infers the rule itself.

Deep learning

ML using neural networks with many layers (“deep”). It powers essentially all of today’s notable AI, including LLMs.

Neural network

A model loosely inspired by brain neurons: layers of simple units connected by weighted links. During training the weights adjust so the network maps inputs to useful outputs. (The neuroscience analogy is loose — artificial “neurons” are far simpler than biological ones.)

Parameters (weights)

The adjustable numbers inside a neural network — what it “learns.” Modern LLMs have billions. More parameters ≠ automatically better.

Generative AI

AI that *produces* new content — text, images, audio, code — rather than only classifying or predicting a label.

Large language model (LLM)

A very large neural network trained on huge amounts of text to predict likely next words. That single ability, at scale, yields writing, summarizing, answering, and reasoning. Claude, GPT, and Gemini are LLMs.

Foundation model

A large model trained on broad data that can be adapted to many downstream tasks — the “base” others build on.

Transformer

The neural-network architecture (introduced 2017) behind modern LLMs. Its key trick is *attention*. The “T” in GPT.

Multimodal

A model that handles more than one kind of input/output — e.g., text *and* images. Claude can read an image of a figure and discuss it.

How models are built & trained

Training

The process of adjusting a model's parameters by showing it data, so its outputs improve. Expensive and done by the model maker — not something you do.

Pretraining

The first, largest training stage: the model learns language and world patterns by predicting next tokens across an enormous text corpus. Produces a capable but unpolished "base model."

Fine-tuning

Additional training on a narrower dataset to specialize or align a model's behavior after pretraining.

RLHF (reinforcement learning from human feedback)

A fine-tuning method where humans rank model outputs and the model is trained to prefer the better ones — a big reason modern assistants feel helpful and follow instructions.

Alignment

The effort to make a model's behavior match human intentions and values — helpful, honest, harmless.

Training data

The text (and images, etc.) a model learned from. Its coverage and biases shape what the model knows and gets wrong.

Knowledge cutoff

The date after which a model has no built-in knowledge, because its training data ended there. Anything newer must come from search or documents you provide.

Benchmark

A standardized test used to compare models (e.g., on reasoning or coding). Useful but imperfect proxies for real-world usefulness.

Model mechanics & internals

Token

The unit of text a model reads and writes — roughly a word-piece (~4 characters of English). Models "think" in tokens; pricing and limits are measured in them.

Context window

How much text (in tokens) a model can consider at once — the prompt plus its reply. Large windows let Claude hold many papers in mind simultaneously.

Embedding

A representation of text as a list of numbers (a vector) that captures meaning, so similar ideas sit near each other in "meaning space." The basis of semantic search.

Inference

Running a trained model to get an output (as opposed to training it). Each chat message is an inference.

Temperature

A setting controlling randomness/creativity in output. Lower = more focused and repeatable; higher = more varied.

Reasoning / “thinking” models

Models that spend extra computation working through a problem step-by-step before answering, improving hard multi-step tasks.

Chain of thought

Prompting or training a model to show intermediate reasoning steps, which often improves accuracy on complex problems.

Using models: prompting & features

Prompt

What you send the model — your instructions, question, and any context. Prompt quality largely determines output quality.

Prompt engineering

The craft of writing effective prompts: giving role, context, constraints, examples, and a desired format.

System prompt

Background instructions that set a model’s persistent behavior for a session (e.g., a Project’s custom instructions).

Zero-shot / few-shot

Asking with no examples (zero-shot) vs. including a few worked examples (few-shot) to steer the output.

Context / grounding

Supplying source material (papers, data, docs) so the model answers *from that* rather than from memory — the key to trustworthy, citable output.

RAG (retrieval-augmented generation)

A pattern where the system first retrieves relevant documents, then has the model answer using them. Claude Projects and NotebookLM are RAG-style tools.

Project (Claude)

A persistent workspace with custom instructions and uploaded knowledge that every chat inside inherits — ideal for a thesis area.

Artifact (Claude)

A live side-panel workspace for a document, table, or small app the model builds and you can edit.

Cowork (Claude)

A desktop feature where Claude runs multi-step tasks over your files and folders in the background.

Canvas / workspace

General term (across tools) for an editable panel where AI-generated content is refined, as opposed to a linear chat.

Agents, tools & connectors

AI agent

A system that doesn't just answer but *acts*: it plans, uses tools, observes results, and loops toward a goal. A chatbot answers; an agent does.

Tool use / function calling

An LLM's ability to call external tools (search, a calculator, a database) to get real information or take actions beyond generating text.

Claude Code

Anthropic's agentic tool that works in a terminal or editor: it reads files, runs commands, and edits — useful for wrangling PDFs, data, and documents, not only for coding.

MCP (Model Context Protocol)

An open standard — a "USB-C port for AI" — that lets Claude connect to outside data and tools through a common plug instead of custom wiring.

Connector

A one-click MCP integration linking Claude to a service — e.g., PubMed, Google Drive, BioRender, Canva.

API (application programming interface)

A way for software to talk to a model programmatically, billed per token. Meant for building applications; a subscription is the right choice for an individual.

Terminal / command line

A text interface for giving a computer typed commands. Basic familiarity is enough to run Claude Code; deep expertise isn't needed.

Guardrails

Constraints that limit what an agent may do — important because an agent's mistake is a performed *action*, not just a wrong sentence.

Research-specific

Grounded search tool

A research tool (Elicit, Consensus, Scite, Semantic Scholar) that answers from a database of *real* indexed papers, so citations correspond to actual sources — far safer than a chatbot's memory.

Smart citations (Scite)

Citation data showing whether later papers *support, mention, or contrast* a finding — useful for checking whether a result held up.

Citation network

The web of which papers cite which; tools like ResearchRabbit map it visually to reveal a subfield.

Reference manager

Software (e.g., Zotero) that stores papers and generates formatted citations. Because it holds only papers you actually added, it's a natural gatekeeper against fake references.

Systematic review

A structured, comprehensive literature review with explicit search and screening criteria; several AI tools now assist the screening/extraction steps.

NotebookLM

Google's free tool that answers strictly from sources you upload, citing each claim — strong for close-reading a fixed set of papers.

Safety, ethics & limitations

Hallucination

When a model produces confident, fluent output that is false — invented facts, fake citations, wrong numbers. A structural property of how LLMs work, not an occasional bug: always verify.

Confabulation

Another word for hallucination, borrowed from the neuroscience/psychology sense of filling gaps with plausible fabrication.

Bias

Systematic skew in outputs reflecting biases in training data — relevant to fairness and to scientific interpretation.

Sycophancy

A model's tendency to agree with the user or tell them what they want to hear — dangerous when you ask leading analytical or statistical questions.

Data privacy / retention

Whether and how a service stores or trains on your inputs. Check settings before pasting unpublished or sensitive data; know your account's controls.

Disclosure

Stating how AI was used in a manuscript or grant, as journals and funders increasingly require.

Research integrity

Honesty and rigor in research. With AI this means: verify every fact and citation, never fabricate data or imagery, and follow authorship, disclosure, and peer-review rules.

Reproducibility

Whether others can repeat your work and get the same result. Document AI-assisted analysis steps so they can be re-run and checked.

Open weights / open source

Models whose parameters are publicly released (e.g., some Llama, Mistral models), which can run locally — relevant when data can't leave your machine.

AGI (artificial general intelligence)

A hypothetical AI with broad, human-level competence across most tasks. Today's systems are narrow by comparison; the term is used loosely — treat bold claims skeptically.

SECTION 7 · DELIVERABLE 3A

Resources used in this plan

Every resource the plan actually assigns, grouped by phase, each checked live in July 2026. The **Link** column is clickable. Costs are the individual/academic rate where one exists. See Section 10 for verification caveats.

Foundations — what AI & LLMs are (Phase 1)

RESOURCE	FORMAT	COST	LINK
3Blue1Brown — Neural Networks series (incl. “But what is a neural network?”, “LLMs explained briefly”, transformers & attention)	Video	Free	3blue1brown.com/topics/neural-networks
Karpathy — “[1hr Talk] Intro to Large Language Models”	Video ~1h	Free	youtube.com/watch?v=zjkBMFhNj_g
Karpathy — “Deep Dive into LLMs like ChatGPT” (reference/deep)	Video 3.5h	Free	youtube.com/watch?v=7xTGNNLPyMI
Lee & Trott — “Large language models, explained with a minimum of math and jargon”	Article	Free	understandingai.org/p/large-language-models-explained
Elements of AI — “Introduction to AI”	Course ~25h	Free	elementsofai.com
DeepLearning.AI — “Generative AI for Everyone” (Andrew Ng)	Course ~6h	Free audit	deeplearning.ai/courses/generative-ai-for-everyone
Ethan Mollick — *Co-Intelligence* (also used in Phase 8)	Book	~$18	penguinrandomhouse.com/books/741805

Claude fluency & prompting (Phase 2)

RESOURCE	FORMAT	COST	LINK
Anthropic — Prompt Engineering Interactive Tutorial	Course (GitHub)	Free	github.com/anthropics/courses
Anthropic Academy — courses catalog (prompting, Claude, Claude Code)	Courses	Free	anthropic.skilljar.com · anthropic.com/learn
Anthropic — Prompt engineering docs	Docs	Free	platform.claude.com/docs
Karpathy — “How I Use LLMs”	Video 2h	Free	youtube.com/watch?v=EWvNQjAaOHw

Literature review & synthesis (Phase 3)

TOOL / RESOURCE	ROLE	COST	LINK

Zotero	Reference manager & citation gatekeeper	Free	**zotero.org**
Semantic Scholar	Free search backbone, TLDRs, alerts	Free	**semanticscholar.org**
ResearchRabbit	Citation-network mapping; Zotero sync	Free	**researchrabbit.ai**
Elicit	Evidence tables from real papers	Free / ~$12/mo	**elicit.com**
Consensus	Citation-backed yes/no claim checks	Free / ~$10/mo	**consensus.app**
Scite	Supporting vs contrasting citations	Library / ~$12/mo	**scite.ai**
NotebookLM	Grounded close-reading + audio overviews	Free	**notebooklm.google**
Retraction Watch (+ database)	Screen references for retractions	Free	**retractionwatch.com**
Tutorials: Elicit systematic-review guide; NotebookLM lit-review guide; ResearchRabbit guide	Tutorials	Free	**elicit.com/blog/systematic-review** · **video**

Data analysis without coding (Phase 4)

RESOURCE	FORMAT	COST	LINK
Anthropic — "Use Claude for Excel"	Official guide	In Pro	**support.claude.com/…/12650343**
OpenAI — "Data analysis with ChatGPT" (optional reference)	Official doc	Feature ~$20/mo	**help.openai.com/…/8437071**
Coupler.io — "Use Claude.ai for Data Analytics (Safely)"	Article	Free	**blog.coupler.io/how-to-use-claude-ai-for-data-analytics**
VelvetShark — "Data analysis with Claude Code" (large files)	Article	Free	**velvetshark.com/data-analysis-with-claude-code**

Scientific writing & research integrity (Phase 5)

RESOURCE	TYPE	COST	LINK
ICMJE Recommendations (AI, authorship, disclosure, peer review)	Policy	Free	**icmje.org/recommendations**
NIH — Generative AI prohibited in peer review (NOT-OD-23-149)	Policy	Free	**grants.nih.gov/…/NOT-OD-23-149**
NIH — "Apply Responsibly" AI-in-applications policy (2025)	Policy	Free	**grants.nih.gov → Apply Responsibly**

ACS Nano — "Best Practices for Using AI When Writing Scientific Manuscripts"	Editorial	Free	pubs.acs.org/doi/10.1021/acsnano.3c01544
MDPI — "Good Practices for Scientific Article Writing with… AI"	Article	Free	mdpi.com/2673-687X/3/2/9

Figures, slides & media (Phase 6)

TOOL / RESOURCE	USE	COST	LINK
BioRender (+ Learning Hub)	Scientific figures & schematics	~$35/mo academic	biorender.com · /learn
draw.io / Excalidraw	Free flowcharts & sketches	Free	drawio.com · excalidraw.com
Google Gemini (image generation)	Concept/slide graphics	Free tier	gemini.google.com
Gamma (+ tutorial)	AI slide decks	Free / ~$8–10/mo	gamma.app
Canva (+ Canva–Claude MCP connector)	Posters, slides, design	Free / Education	canva.com · /help/mcp-agent-setup

Claude Code, agents & connectors (Phase 7)

RESOURCE	LEVEL	COST	LINK
Anthropic — Terminal guide for new users	Absolute beginner	Free	code.claude.com/docs/en/terminal-guide
MDN — Command line crash course (§1–3)	Beginner	Free	developer.mozilla.org → Command line
IBM Technology — "What are AI Agents?"	Beginner video	Free	youtube.com/watch?v=F8NKVhkZZWI
explainX — "What Are AI Agents? A Plain-English Beginner's Guide (2026)"	Beginner	Free	explainx.ai/blog/what-are-ai-agents…
Anthropic — Claude Code Quickstart	Beginner	Free	code.claude.com/docs/en/quickstart
Anthropic Academy — "Claude Code 101" → "Claude Code in Action"	Beginner → Int.	Free	anthropic.skilljar.com · Coursera
CC for Everyone — "Claude Code for Non-Developers"	Non-coder	Free	ccforeveryone.com/…/non-developers
CC for Everyone — "Claude Code for Researchers"	Non-coder	Free	ccforeveryone.com/…/researchers
Anthropic — "Introducing MCP" & the MCP intro docs	Beginner → Int.	Free	anthropic.com/news/mcp · modelcontextprotocol.io
Anthropic — "Building effective agents" (reference only; developer-level)	Advanced	Free	anthropic.com/engineering/building-effective-agents

Integration, judgment & responsible use (Phase 8)

RESOURCE	TYPE	COST	LINK
Ethan Mollick — *Co-Intelligence*	Book	~$18	**penguinrandomhouse.com/books/741805**
MIT Sloan — “When AI Gets It Wrong: Hallucinations & Bias”	Article	Free	**mitsloanedtech.mit.edu/…/hallucinations-and-bias**
Narayanan & Kapoor — AI Snake Oil / “AI as Normal Technology”	Book + essays	Free essays	**normaltech.ai**

If you have extra time

Optional depth — genuinely good resources that would over-stuff a 1-hour-a-day plan, but reward extra curiosity. Not required to reach the end state.

Deeper on fundamentals & prompting

- **Karpathy — "Deep Dive into LLMs like ChatGPT"** (3.5h video) — the complete training-stack reference if Phase 1 leaves you wanting more.
- **DeepLearning.AI — "ChatGPT Prompt Engineering for Developers"** (~1.5h, free) — hands-on prompting; assumes a little Python, so optional for a non-coder.
- **Anthropic Courses on GitHub** (API fundamentals, tool use, evaluations) and the Anthropic Cookbook — code-heavy; for if you ever want to peek under the hood.

More literature-review firepower

- **Undermind** — agentic "deep" literature search with near-exhaustive recall on narrow mechanistic questions; worth it during an intensive systematic-review push. Confirm current price at signup.
- **SciSpace** — all-in-one chat-with-PDF + formatting; check for a free university license.
- **Perplexity** **(student ~$10/mo)** — fast web-grounded orientation and background; always click through to sources (it searches the open web, not just papers).
- **Zotero AI plugins** — PapersGPT (chat-with-PDF in your library; supports free local models), Aria, Beaver.

More image / media tools

- **Ideogram** (best text-in-image, free tier) for poster titles/labels; **Adobe Firefly** (commercially-safe training data) when licensing matters; **Midjourney** for top-tier aesthetics (paid, overkill for most needs).
- **AI video (free tiers):** Google Veo via Gemini; Pika and Kling for cheap experimentation; Runway/HeyGen only if a specific project demands it. Video is a nice-to-have, not core.

Worth checking with the institution

- **Claude for Education** — if Vanderbilt participates, it could make paid Claude access free. **Scite** and **Microsoft 365 Copilot** may also be library/IT-provided. Five minutes could save you real money.

Continuing education — the keep-current habit

This field moves monthly; staying current is a *light* habit, not a second job. The list is deliberately ruthless: **two podcasts and two newsletters** is the whole recommendation. Everything else is noise for a busy scientist.

Podcasts — pick 2

SHOW	WHY IT'S THE PICK	CADENCE	LINK
Hard Fork (NYT)	#1 pick. Weekly, journalistic, entertaining, jargon-free — the best "keep up without drowning" show for a non-coder.	Weekly	**nytimes.com/column/hard-fork**
Google DeepMind: The Podcast (Hannah Fry)	Best science fit. A mathematician interviewing researchers on AI in science, health, and biology — matches your domain and level.	Fortnightly (by season)	**deepmind.google/the-podcast**
Optional 3rd: The AI Daily Brief	~20-min daily current-events drip if you want more — easy to skip when the hour is tight.	Daily	**Apple Podcasts**

Cut for fit: Latent Space & Practical AI (too engineering-focused), Dwarkesh (superb but long/dense — a "graduate" option later), No Priors (VC framing), Lenny's (product management). Watch for Latent Space's spun-off "AI for Science" show — promising domain fit, not yet verified for cadence.

Newsletters — pick 2–3 (all free)

NEWSLETTER	WHY IT'S THE PICK	LINK
One Useful Thing (Ethan Mollick)	#1 pick. Practical, non-technical, experiment-driven — the best ongoing "how do I actually use this" source.	**oneusefulthing.org**
The Batch (DeepLearning.AI)	Weekly, well-curated by Andrew Ng's team — balanced signal without hype.	**deeplearning.ai/the-batch**
AI as Normal Technology (AI Snake Oil)	The skeptical, research-integrity voice — occasional deep reads, not weekly. The right counterweight for a scientist.	**normaltech.ai**

Optional level-ups: Simon Willison's weblog (excellent, developer-leaning) and Import AI by Jack Clark (research-dense). Skip TLDR AI — high-volume, tooling-heavy.

YouTube — optional, pick at most 2

- **Google DeepMind** — research explainers + video podcast, on-domain.
- **AI Explained** — clear, analytical, non-hype breakdowns of why a model or trend matters.
- *Alternative:* **Two Minute Papers** for short, upbeat research summaries.

THE HABIT, CONCRETELY

One podcast on the commute/at the bench, one newsletter over coffee each week. That’s it. If a tool or model you use ships a major update, spend one session trying it on real work — hands-on beats reading about it.

SECTION 10

Verification notes & caveats

Honesty about sourcing. This plan was researched by parallel specialist passes, each required to confirm links resolve and content matches before inclusion. Here's what was solid and what to re-check.

Method

Six research streams (Anthropic ecosystem; AI/LLM fundamentals; literature tools; Claude Code for non-coders; image/video/figure tools; writing/productivity/continuing-ed) were run and cross-checked against the live web on 8–9 July 2026. Two official Anthropic documentation pages were loaded in full; most other pages were confirmed via search results returning the page's own current text. Nothing dead, dormant, or unverifiable was knowingly included.

Known limitations — re-check these before relying on a number

- **Vendor bot-blocking.** Several sites (anthropic.com, claude.com, YouTube, BioRender, Canva, some Substacks) return errors to automated fetchers. Their facts here are corroborated from official-page search snippets plus independent sources — open normally to confirm the exact figure.
- **Consumer-AI pricing shifts monthly.** All prices are July-2026 rates; confirm at signup, ideally with a .edu email for academic discounts (Elicit, Consensus, BioRender, Perplexity, Scite all offer them — exact post-discount numbers vary).
- **Claude plan → model mapping.** Which specific model each tier includes changes as Anthropic ships; the safe statement is "Pro/Max include the current Opus/Sonnet/Haiku families."
- **Claude Science workbench.** Details (macOS/Linux beta, plan availability, any academic-lab discount) came from announcement snippets, not a loaded page. It isn't needed for this plan regardless.
- **Undermind pricing** and the **Latent Space "AI for Science" podcast** were flagged by research as not fully verifiable — both are optional (Section 8/9), so this doesn't affect the core path.
- **NIH / ICMJE policy is living guidance.** Always check the current version before a specific submission — funder and journal AI rules are still evolving.

TWO THINGS THAT DON'T CHANGE

Whatever the prices do: (1) **never let any chatbot invent a citation** — verify every reference against a real source; and (2) **never generate imagery that could pass as experimental data.** These are the load-bearing integrity rules of the entire plan.

APPENDIX A

First-week checklist & starter prompts

Week-one setup checklist

- ✓ Subscribe to **Claude Pro**; install the desktop app; open claude.ai.
- ✓ Review privacy/memory/training settings and set them to your comfort.
- ✓ Confirm you have a **Google account** (for NotebookLM & Gemini later).
- ✓ Create your **AI Lab Notebook** (one doc for prompts, wins, and errors).
- ✓ Bookmark **Anthropic Academy** and the Section 9 podcast + newsletter.
- ✓ Ask your library/IT whether Vanderbilt provides **Claude for Education** or Scite access.
- ✓ Do the first useful chat (below) on a paper you're actually reading.

Starter prompts (copy, then adapt — wet-lab flavored)

UNDERSTAND A PAPER

"I'm a neuroscience PhD student. Here is a paper's abstract [paste]. Explain its central finding in 4 sentences, define any specialized terms, list the main method used, and give two questions a critical reviewer would raise. Don't speculate beyond the text."

SET UP YOUR THESIS PROJECT (CUSTOM INSTRUCTIONS)

"You are assisting a Vanderbilt MD/PhD student doing basic-science neuroscience in a wet lab studying [system/technique]. Prefer precise, cautious, citable answers. When I ask about literature, only use sources I provide — never invent citations. Flag uncertainty explicitly."

SYNTHESIZE PAPERS YOU UPLOADED (GROUNDED)

"Using only the PDFs in this Project, summarize what is known about [topic]. Group by theme, note where papers agree or disagree, and cite the specific source for every claim. List open questions. If something isn't in these papers, say so rather than guessing."

ANALYZE A DATASET (WITH PUSHBACK BUILT IN)

"Here is a CSV [upload]. Describe its structure and any data-quality issues. Propose a cleaning plan and wait for my approval. Then recommend an appropriate statistical test, state the assumptions it requires, and check whether my data meet them — tell me if it's the wrong test rather than just running it."

EDIT YOUR WRITING (YOUR WORDS, YOUR CLAIMS)

“This is a paragraph I wrote for a manuscript [paste]. Improve clarity, flow, and concision without changing my meaning or adding any claims or citations. Keep my voice. Then, separately, tell me where a study-section reviewer would push back.”

Prepared as a personalized mentorship plan for Ezechukwu Nduka · July 2026. Anchored on Anthropic’s Claude, with best-in-class non-Anthropic tools where they win. All resources verified against the live web at time of writing; confirm prices and policies at point of use. The two integrity rules — verify every citation, never fabricate data imagery — are non-negotiable.